

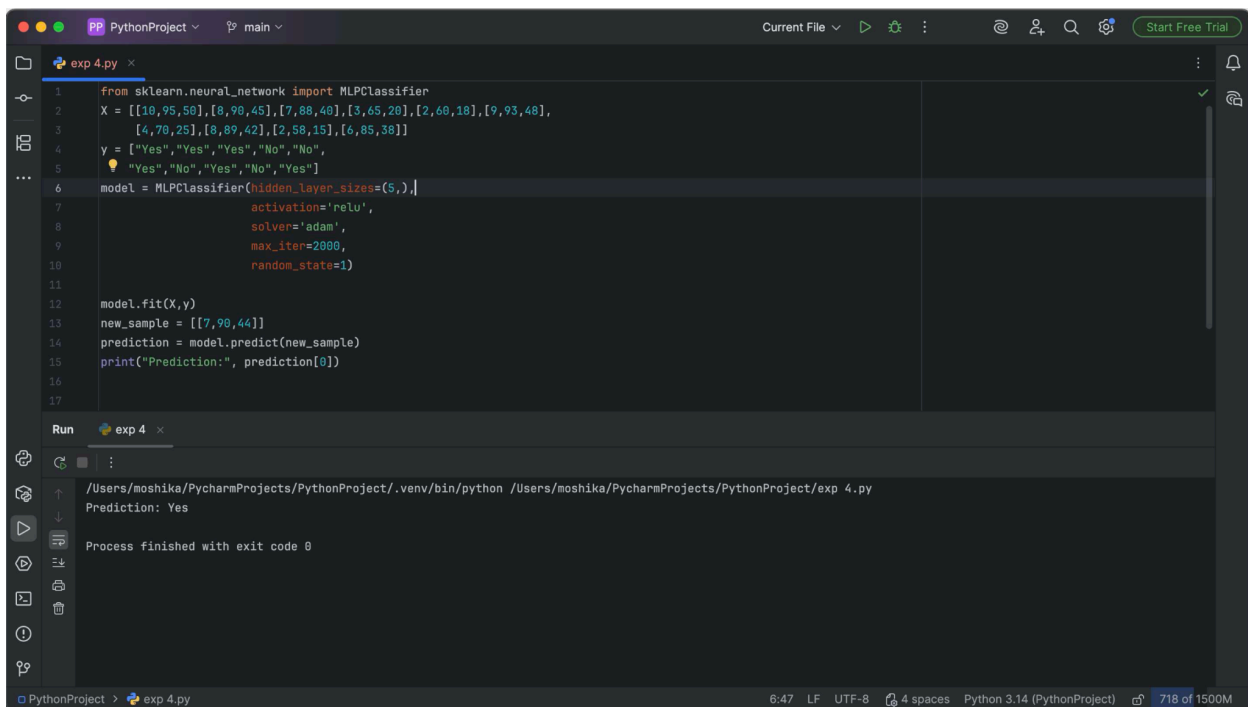
EXPERIMENT - 4

Aim: To implement an Artificial Neural Network using the Backpropagation algorithm and test its performance on a suitable dataset.

Procedure:

1. Load the training dataset.
2. Initialize the network weights and biases.
3. Give the input values to the input layer.
4. Perform forward propagation.
5. Calculate the output error.
6. Propagate the error backward through the network.
7. Update the weights and biases.
8. Repeat the process for multiple epochs.
9. Test the trained network using test data.
10. Display the predicted output and accuracy.

Output:



```
1 from sklearn.neural_network import MLPClassifier
2 X = [[10,95,50],[0,90,45],[7,88,40],[3,65,20],[2,60,10],[9,93,40],
3      [4,70,25],[8,89,42],[2,58,15],[6,85,38]]
4 y = ["Yes","Yes","Yes","No","No",
5      "Yes","No","Yes","No","Yes"]
6 model = MLPClassifier(hidden_layer_sizes=(5,))
7     activation='relu',
8     solver='adam',
9     max_iter=2000,
10    random_state=1)
11
12 model.fit(X,y)
13 new_sample = [[7,90,44]]
14 prediction = model.predict(new_sample)
15 print("Prediction:", prediction[0])
16
17
```

Run exp 4

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 4.py
Prediction: Yes
Process finished with exit code 0
```

PythonProject > exp 4.py 6:47 LF UTF-8 4 spaces Python 3.14 (PythonProject) 718 of 1500M

Result: The Backpropagation algorithm successfully trains the Artificial Neural Network and produces predictions for the test data.